Bidding-Based Online Policy Adaptation for Evolving Objectives

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Summary

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Contribution(s)

1. Provide a succinct but precise list of the contribution(s) of the paper. Use contextual notes to avoid implications of contributions more significant than intended and to clarify and situate the contribution relative to prior work (see the examples below). If there is no additional context, enter "None". Try to keep each contribution to a single sentence, although multiple sentences are allowed when necessary. If using complete sentences, include punctuation. If using a single sentence fragment, you may omit the concluding period. A single contribution can be sufficient, and there is no limit on the number of contributions. Submissions will be judged mostly on the contributions claimed on their cover pages and the evidence provided to support them. Major contributions should not be claimed in the main text if they do not appear on the cover page. Overclaiming can lead to a submission being rejected, so it is important to have well-scoped contribution statements on the cover page.
   Context: None
2. The submission template for submissions to RLJ/RLC 2025
   Context: Built from previous RLC/RLJ, ICLR, and TMLR submission templates
3. *[Example of one contribution and corresponding contextual note for the paper "Policy gradient methods for reinforcement learning with function approximation" (**?**).]*
   This paper presents an expression for the policy gradient when using function approximation to represent the action-value function.
   Context: Prior work established expressions for the policy gradient without function approximation (**?**).

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Abstract

We study multi-objective reinforcement learning settings in which objectives appear or disappear at runtime. We propose a modular framework where each objective is supported by a selfish local policy, and coordination is achieved through a novel *auction*-based mechanism: policies bid for the right to execute their actions, with bids reflecting the urgency of the current state. The highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives. To succeed, each policy must not only optimize its own objective, but also reason about the presence of other goals and learn to produce calibrated bids that reflect relative priority. When objectives change, the system adapts by simply adding or removing the corresponding policies. Moreover, when objectives arise from the same parameterized family—like the class of reachability objectives parameterized by target states—identical copies of a parameterized policy can be deployed. In our implementation, the policies are trained concurrently using proximal policy optimization (PPO). We evaluate on Atari Assault and a gridworld-based path-planning task with dynamic targets. Our method achieves substantially better performance and reduced sample complexity than a single policy trained with PPO.

1 Introduction

A large class of real-world control problems requires the simultaneous fulfillment of multiple objectives, which may be partially conflicting. We study a setting in which objectives can appear or disappear during deployment, without any prior information about their arrival process. As a running example, consider a table-serving robot in a coffee shop. Each coffee request is modeled as a discounted payoff objective. New requests may arise at arbitrary locations and times, while existing requests may be canceled before being served. At any moment, an arbitrary number of requests may be active, and no assumptions are made about their distribution. The robot's policy must therefore adapt dynamically to achieve an optimal trade-off among the payoffs associated with the currently active requests. Our goal is to design lightweight, adaptive policies that update their behavior seamlessly as objectives change.

This problem is of significant practical importance and has been extensively studied in the planning literature (**?**). However, most existing approaches assume a precise environment model and the absence of uncertainty. In contrast, within reinforcement learning (RL) settings—where the environment may be uncertain and a model may not be available—this problem has received little attention.

We propose a novel policy adaptation algorithm tailored to this class of problems. The central idea is a compositional design for multi-objective RL, in which each objective is supported by an independent policy, and coordination is achieved through an online auction-based mechanism. At every decision step, policies submit bids for the right to execute their actions, where bids reflect the

urgency of the current state with respect to their objective. T he highest bidder selects the action, enabling a dynamic and interpretable trade-off among objectives.

Returning to the coffee shop example, suppose the robot approaches an intersection with two active requests. The policy for the first request recommends turning left, while the policy for the second recommends turning right. Since both actions cannot be executed simultaneously, a trade-off is unavoidable. The auction mechanism determines which objective is prioritized, based on the relative urgency encoded in the bids.

This compositional design yields strong modularity. When an objective appears or disappears, only the corresponding policy needs to be added or removed, without modifying the remainder of the system. Furthermore, when objectives belong to the same parameterized family—for example, coffee requests parameterized by their locations—a single universal policy can be trained for the entire family. New objectives can then be instantiated as identical copies of this policy, enabling immediate adaptation. In contrast, a monolithic policy serving all objectives lacks this flexibility. Training such a policy under a variable number of objectives leads to substantial degradation in both training stability and performance, as we demonstrate empirically.

We show how classical RL policies can be augmented with bidding capabilities. A natural approach is to extend the action space with numeric bid values; however, this introduces three key challenges.

Challenge I: Enforcing honest bids. Policies must bid in proportion to their true urgency. Otherwise, they may learn to overbid systematically in order to suppress competitors, resulting in dysfunctional behavior. To prevent this, we impose penalties during training that are proportional to the bid values. **KM:** I suggest, we just stick to one kind of bidding, instead of three, if in the end all-pay remains superior in all experiments. As a consequence, excessive bidding reduces the net return of a policy, aligning its incentives with truthful urgency estimation.

Challenge II: Achieving environment awareness. Effective bidding depends not only on the current state and the local objective, but also on the configuration of competing objectives. For example, if two requests are spatially close, their policies need not compete aggressively. Conversely, if requests lie in opposite directions, the more urgent policy must bid sufficiently high to prevail. To enable such environment-aware bidding, we formulate local policy learning as a multi-agent nonzero-sum game, where each agent represents an objective and learns to optimize its payoff in the presence of competing agents.

Challenge III: Unbounded objective count. During deployment, the number of active objectives may exceed those observed during training. This invalidates standard game-theoretic training formulations, which assume a fixed number of opponents. To address this issue, we employ an attention mechanism that aggregates an arbitrary number of competing objectives into a fixed-dimensional representation. This encoding is then provided as input to the policy network. Each local policy is equipped with its own attention module, co-trained with the policy across progressively increasing numbers of objectives. To ensure scalability, we focus on objectives drawn from a common parameterized family, which allows shared structure across instances. **KM:** If we want to make the unbounded objectives a central feature, we should restrict ourselves to objectives from the same family. Otherwise, the attention pooling and the game-theoretic training approach does not work. While this is indeed a restrictive case, it is well-motivated and I do not see it as a weakness.

We implement our framework using proximal policy optimization (PPO) as the base learning algorithm. On the coffee-serving robot benchmark, modeled using a gridworld-based path-planning problem, and on the Atari Assault benchmark, our approach demonstrates significantly improved training stability and superior performance compared to monolithic baselines. In particular, compositional training converges faster and exhibits substantially lower variance in returns, highlighting the practical benefits of modular auction-based coordination. **TODO:** Summarize the main take-away: can we give some statistics, like “the training time was 2x faster” or the “loss was 2x smaller”?

1.1 Related Work

A large body of work in multi-objective reinforcement learning (MORL) relies on *scalarization*, aggregating multiple reward functions into a single scalar objective so that standard single-objective RL algorithms can be applied. The simplest scalarization method is a weighted sum of individual rewards (Gass & Saaty, 1955), though richer nonlinear scalarization functions have also been proposed (Van Moffaert et al., 2013). A key limitation of scalarization is that the relative importance induced by the aggregation function may not align with the designer's true intent. This mismatch can initiate a tedious debugging cycle, particularly in large-scale systems (Hayes et al., 2022). In contrast, our approach achieves a trade-off between reward components without collapsing them into a fixed scalar objective.

Other works pursue trade-offs by fixing a specific optimality criterion. Common choices include Pareto optimality (Van Moffaert & Nowé, 2014) and its approximations (Pirotta et al., 2015), as well as fairness-based criteria across reward functions (Park et al., 2024; Byeon et al., 2025; Siddique et al., 2020). These approaches typically learn a single monolithic policy that satisfies the chosen criterion. By contrast, our objective is to learn independent, selfish local policies for each reward component and compose them at runtime in a principled manner, thereby preserving modularity while still achieving a coherent global trade-off.

Relatively few works study distributed local policies for multiple rewards. A notable example is W-learning (Humphrys, 1995) and its deep RL extension (Rosero et al., 2024), where separate selfish policies are trained alongside meta-policies (W-functions) that assign each state a score reflecting its urgency. At runtime, the policy with the highest score is selected. Other approaches employ alternative aggregation mechanisms, such as ranked voting over actions (Méndez-Hernández et al., 2019), or fixed aggregation rules like summing action values across agents (Russell & Zimdars, 2003). While conceptually related, our approach is technically simpler: it relies on an engineered reward structure that enables the use of standard learning algorithms (e.g., PPO) without additional meta-policies or complex aggregation schemes. Furthermore, to the best of our knowledge, we are the first to introduce the incremental MORL setting, in which reward components can be added or removed at runtime.

The idea of bidding-based selfish policies originates from analogous techniques for multi-objective path planning problems on finite graphs (Avni et al., 2024), as well as from the broader literature on bidding games (Lazarus et al., 1999; Avni et al., 2019; 2025). These works study strategic interaction in finite arenas, where adversarial players bid for the right to determine the next move from a shared action space in pursuit of their objectives. Although these works provide strong theoretical guarantees, they do not naturally extend to infinite arenas. Moreover, players in such games are typically budget-constrained, and the central question concerns the minimum budget required to win. In contrast, we consider infinite arenas and eliminate explicit budget constraints by incorporating bidding rewards and penalties directly into the learning framework.

2 Preliminaries: Multi-Objective MDPs

Mainstream RL algorithms consider Markov decision processes (MDP) equipped with a *single* reward function, pertaining to a single task or *objective* for the system. In reality, a majority of real-world applications of RL requires satisfying multiple, partly contradictory objectives. We model such multi-objective decision-making problems using multi-objective MDPs (MO-MDP), as formally defined below. Intuitively, an MO-MDP has the exact same syntax as a regular MDP, except that it now has multiple reward functions pertaining to the different objectives. We formalize MO-MDP below. We will use the notation $\mathbb{D}(\Sigma)$ to represent the set of all probability distributions over a given alphabet Σ.

Definition 1 (MO-MDP)**.** A multi-objective Markov decision process (MO-MDP) with $m \in \mathbb{Z}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, \mathbf{R}, \mu_0)$, where

- S is the set of states,

- A is the set of actions,
- $T : S \times A \to \mathbb{D}(S)$ is the transition function mapping a state-action pair to a distribution over the successor states,
- $\mathbf{R} = \{R^i : S \times A \times S \to \mathbb{R}_{\geq 0}\}_{i \in [1;m]}$ is the set of reward functions, and
- $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution.

The notions of policies and paths induced by them are exactly the same as in classical MDPs, which we briefly recall below. First, we introduce some notation. Given an alphabet Σ, we will write Σ^* and Σ^ω to denote the set of every finite and infinite word over Σ, respectively, and will write $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$. Given a word $w = \sigma_0\sigma_1 \ldots \in \Sigma^\infty$, and given a $t \geq 0$ that is not larger than the length of w, we will write w_t and $w_{0:t}$ to denote respectively the t-th element of w, i.e., $w_t = \sigma_t$, and the prefix of w up to the t-th element, i.e., $w_{0:t} = \sigma_0 \ldots \sigma_t$.

A *policy* in an MO-MDP $\mathcal{M}$ is a function $\pi : (S \times A)^* \times S \to \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. A *path* on $\mathcal{M}$ induced by π is a sequence $\rho = (s_0, a_0)(s_1, a_1), \ldots \in (S \times A)^\infty$ such that for every $t \geq 0$, (1) the probability that the action a_{t+1} is picked by π based on the history is positive, i.e., $\pi(\rho_{0:t}, s_{t+1})(a_{t+1}) > 0$, and (2) the probability of moving to the state s_{t+1} from s_t due to action a_t is positive, i.e., $T(s_t, a_t)(s_{t+1}) > 0$. A path can be either finite or infinite, and we will write $Paths(\mathcal{M}, \pi)$ to denote the set of all infinite paths fo $\mathcal{M}$ induced by π. Given a finite path $\rho = (s_0, a_0) \ldots (s_t, a_t)$, the probability that ρ occurs is given by: $\mu_0(s_0) \cdot \prod_{k=0}^{t-1} T(s_k, a_k)(s_{k+1}) \cdot \pi(\rho_{0:k}, s_{k+1})(a_{k+1})$. This can be extended to a probability measure over the set of all infinite paths in $\mathcal{M}$ using standard constructions, which can be found in the literature (Baier & Katoen, 2008). Given a measurable set of paths Ω and a function $f \colon Paths(\mathcal{M}, \pi) \to \mathbb{R}$, we will write $\mathbb{P}^{\mathcal{M},\pi}[\Omega]$ and $\mathbb{E}^{\mathcal{M},\pi}[f]$ to denote, respectively, the probability measure of Ω and the expected value of f evaluated over random infinite paths.

We will use the standard discounted reward objectives, where we fix $\gamma \in (0, 1)$ as a given discounting factor. Let $\rho = (s_0, a_0)(s_1, a_1), \ldots \in Paths(\mathcal{M}, \pi)$ be an infinite path induced by π. Define the discounted sum function, mapping ρ to the discounted sum of the associated rewards: $f^i_{\text{ds}}(\rho) := \sum_{t=0}^{\infty} \gamma^t \cdot R^i(s_t, a_t)$. The *$i$-value* of the policy ρ for $\mathcal{M}$ is the expected value of the discounted sum of the i-th reward we can secure by executing ρ on $\mathcal{M}$, written as $val^{\mathcal{M},i}(\pi) = \mathbb{E}^{\mathcal{M},\pi}[f^i_{\text{ds}}]$. The *optimal* policy for R^i for a given $i \in [1; m]$ is the policy that maximizes the i-value. When the reward index i is unimportant, we will refer to every element of the set $\{val^{\mathcal{M},i}\}_{i \in [1;m]}$ as a *value component*.

When the MO-MDP $\mathcal{M}$ is clear from the context, we will drop it from all notation and will simply write $Paths(\pi)$, $\mathbb{P}^\pi$, $\mathbb{E}^\pi$, and val^i.

It is known that *memoryless* (aka, stationary) policies suffice for maximizing single discounted reward objectives, where a policy π is called memoryless if the proposed action only depend on the current state. In other words, given every pair of finite paths ρ, ρ' both ending at the same state, the probability distributions $\pi(\rho)$ and $\pi(\rho')$ are identical.

Unlike classical single-objective MDPs, the optimal policy synthesis problem for MO-MDP requires fixing one of many possible optimality criteria. Many possibilities exist, including pareto optimality, requiring a solution where none of the value components could be unanimously improved without hurting the others; weighted social welfare, requiring a weighted sum of the value components be maximized; and fairness, requiring the minimum attained value by any value component is maximized. **TODO:** Give some citations for each category.

3 Auction-Based Compositional RL on Multi-Objective MDPs

We consider the compositional approach to policy synthesis for MO-MDPs, where we will design a selfish, *local* policy maximizing each individual value component, towards the fulfillment of some required global coordination requirements. The main crux is in the composition process, where

each local policy may propose a different action, but the composition must decide one of the actions that will be actually executed. Importantly, the composition must be implementable in a distributed manner, meaning we will *not* use any global policy that would pick an action by analyzing all local policies and their reward functions. **TODO:** running example

3.1 The Framework

We present a novel *auction*-based RL framework for compositional policy synthesis for MO-MDPs. In our framework, not only do the local policies emit actions, but also they *bid* for the privilege of executing their actions for a given number of time steps $\tau \in \mathbb{N}_{>0}$ in future. The bids are all non-negative real numbers, and the highest bidder's actions get executed for the following τ consecutive steps, with bidding ties being resolved uniformly at random. The policy whose actions are executed is referred to as the *winning* policy, and it must pay a bidding *penalty* that equals to its bid amount; this is to discourage overbidding. The policies whose actions are not executed are called the *losing* policies, and we consider three different settings for the "payment" they must make:

Loser-Rewarded: the winning policy pays the bidding penalty and the losing policies earn bidding rewards equal to their respective bid values;

Winner-Pays: the winning policy pays the bidding penalty and the losing policies are unaffected (i.e., neither earn bidding rewards nor pay bidding penalties);

All-Pay: all policies pay bidding penalties equal to their respective bid values.

While penalizing the winner discourages overbidding, the situation with the losers is more subtle. In the Loser-Rewarded setting, by rewarding the losers, we encourage policies to bid positively if the current state has some importance to them; this way, if they lose the bidding, they will get some positive reward. In the All-Pay setting, by penalizing all policies, we discourage policies to bid at all unless it is absolutely important. The Winner-Pays setting balances these two: by neither rewarding nor penalizing the losers, we neither encourage nor discourage policies to bid. In Section 3.3, we will see how these three settings induce different kinds of coordination through bidding.

For each policy, the bidding penalty or reward gets, respectively, subtracted or added to the *nominal* reward obtained from the reward functions of the given MO-MDP, and the resulting reward is called the *net* reward.

In summary, through this novel bidding mechanism, each policy can adjust its bid in proportion to the importance for it to execute its action in the current state, and the associated bidding penalty/reward aims to incentivize policies to be truthful. By making the highest bidder active, it is effectively guaranteed that the most important policy is executed. This way, we obtain a purely decentralized scheme to coordinate local policies in a given MO-MDP.

Remark 1 (On the parameter τ). The parameter τ controls how frequently the agent changes its policies. In practice, if τ is too small, the switching could be too frequent for any of the objectives to be fulfilled. For example, **TODO:** running example...

3.2 The Design Problem and Learning Algorithms

We consider the following learning task for our auction-based compositional framework:

Given an MO-MDP, a constant $\tau > 0$, and $\Delta \in \{$Loser-Rewarded, Winner-Pays, All-Pay$\}$, compute local policies that are optimal for the net rewards obtained in the mode Δ, given that all other local policies behave selfishly towards maximizing their own net rewards.

We will show how the above learning problem boils down to solving a standard learning problem in the multi-agent setting, formalized using a decentralized MDP (DEC-MDP) as defined below. The only difference between a DEC-MDP and an MO-MDP (see Definition 1) is that now each reward function R^i is owned by the Agent i, who now controls a separate set of actions A^i.

Definition 2 (DEC-MDP)**.** A decentralized Markov decision process (DEC-MDP) with $m \in \mathbb{Z}_{>0}$ agents is specified by a tuple $\mathcal{M} = (S, \mathbf{A}, T, \mathbf{R}, \mu_0)$, where

- S is the set of states,
- $\mathbf{A} = \{A^1, \ldots, A^m\}$ is a set with A^i being the set of Agent i's actions,
- $T : S \times A^1 \times \ldots \times A^m \to \mathbb{D}(S)$ is the transition function mapping a state-action pair to a distribution over the successor states,
- $\mathbf{R} = \{R^i \colon S \times A^1 \times \ldots \times A^m \times S \to \mathbb{R}_{\geq 0}\}_{i \in [1;m]}$ is the set of reward functions, and
- $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution.

The definitions of policies and paths readily extend from MO-MDP to DEC-MDP.

Given a DEC-MDP, the goal is to compute an ensemble of local (memoryless) policies for all individual agents, such that for every $i \in [1; m]$, the i-value cannot be increased by a unanimous change of the local policy π^i. In other words, the goal is to find a set of selfish local policies that are in a Nash equilibrium. This is an extensively studied problem in the literature. **TODO:** Do a little bit of literature survey...

Our focus is not in improved algorithms for DEC-MDP, but rather to show how the local policy synthesis problem for the MO-MDP $\mathcal{M}$ in our auction-based framework reduces to the multi-agent policy synthesis problem in a DEC-MDP $\widetilde{\mathcal{M}}$. Intuitively, for every state s of $\mathcal{M}$, $\widetilde{\mathcal{M}}$ creates two kinds of copies, ones where bidding happens and are represented simply as s, and ones of the form (s, t, i^*) that keeps track of the time t elapsed since the last bidding, and the winner i^* of the last bidding. Furthermore, bidding is facilitated by extending the action space of $\mathcal{M}$ to include all real-valued bids, and each agent in $\widetilde{\mathcal{M}}$ has an identical copy of this extended action space. After bidding in a state s, the winner i^* is selected, and the state moves to $(s, 0, i^*)$. From this point onward, only Agent i^* selects actions $a^0, a^1, \ldots, a^\tau$ to produce the sequence $(s^1, 1, i^*), (s^2, 2, i^*), \ldots, (s^{\tau-1}, \tau - 1, i^*), s^\tau$, after which the next bidding happens, and the process repeats. Finally, the bidding penalties or bidding rewards are only paid during the transition $s \to (s, 0, i^*)$, otherwise, the rewards are inherited from the original MO-MDP.

We formalize this below. Given an MO-MDP $\mathcal{M} = (S, A, T, \mathbf{R}, \mu_0)$, a constant $\tau > 0$, and the mode $\Delta \in \{\textsf{Loser-Rewarded}, \textsf{Winner-Pays}, \textsf{All-Pay}\}$, we define the DEC-MDP $\widetilde{\mathcal{M}} = (\widetilde{S}, \widetilde{\mathbf{A}}, \widetilde{T}, \widetilde{\mathbf{R}}, \widetilde{\mu}_0)$ where

- $\widetilde{S} \coloneqq S \cup S \times [0; \tau - 1] \times [1; m]$,
- $\widetilde{\mathbf{A}} \coloneqq \{\widetilde{A}^i\}_{i \in [1;m]}$ where $\widetilde{A}^i \coloneqq A \cup \mathbb{R}$,
- $\widetilde{\mu}_0 \coloneqq \mu_0 \times \{0\}$,

and for every current state $s \in \widetilde{S}$ and every current action $(b^1, \ldots, b^m) \in \mathbb{R}^m$, writing the highest bidders as $I = \{i \in [1; m] \mid \forall j \in [1; m] \,.\, b^i \geq b^j\}$,

- $\widetilde{T}(s, b^1, \ldots, b^m) \coloneqq \text{Uniform}(\{(s, 0, i)\}_{i \in I})$,
- $\widetilde{R}^i(s, b^1, \ldots, b^m, (s, 0, i^*)) \coloneqq \begin{cases} -b^i & i = i^* \vee \Delta = \textsf{All-Pay}, \\ +b^i & i \neq i^* \wedge \Delta = \textsf{Loser-Rewarded}, \\ 0 & i \neq i^* \wedge \Delta = \textsf{Winner-Pays}, \end{cases}$

whereas if the current state is of the form $(s, t, i^*) \in \widetilde{S}$, for every action $(a^1, \ldots, a^m) \in A^m$,

- $\widetilde{T}((s, t, i^*), a^1, \ldots, a^m) \coloneqq \begin{cases} T(s, a^{i^*}) \times ((t+1) \mod \tau) \times \{i^*\} & t < \tau - 1, \\ T(s, a^{i^*}) & t = \tau - 1, \end{cases}$
- $\widetilde{R}^i((s, t, i^*), a^1, \ldots, a^m, (s', t+1, i^*)) \coloneqq R^i(s, a^i, s')$.

KM: A soundness theorem would be good, but what can we say concretely?

3.3 Flavors of Cooperation through Bidding

We provide theoretical insights into the global behavior that emerges out of the auction-based interactions between the local policies. For the sake of theoretical guarantees, and to be able to convey the main essence of our results, we choose the simplest bare bone setting:

Assumption 1. The given MO-MDP has finite state and action spaces, and for every (memoryless) policy, the bottom strongly connected component (BSCC) of the resulting Markov chain (MC) is a sink state where no reward is earned. Furthermore, the time parameter $\tau = 1$, meaning the bidding takes place at each time step before selecting the action.

Firstly, since the MO-MDP is finite, for each individual reward function, *deterministic* memoryless policy suffices. **TODO:** give some citation

The following two types of global behaviors are of particular interest:

Social welfare is the sum (equivalently, the average) of the i-values for all i. We may ask: is the emergent global behavior guaranteed to achieve the maximal social welfare?

Fairness is measured by the disparity between different i-values, i.e., $\max_{i,j\in[1;m]} |val^i - val^j|$. Fairness is maximized when the disparity is minimized. We may ask: is the emergent global behavior guaranteed to achieve the maximal fairness?

Theorem 1. *Suppose the MO-MDP is such that at each state s and for every action a, there exists at most a single $i \in [1;m]$ such that the optimal policy for R^i selects a at s. Then, the* Loser-Rewarded *setting maximizes the social welfare.*

Proof sketch. First, consider the simple one-shot game, where the agents bid just one time to select an action, and the reward is based on the resulting single probabilistic transition. Suppose for the index $i \in [1;m]$, the expected reward from using the action $a \in A$ is E^i_a, and define $E^i_+ := \max_{a\in A} E^i_a$ and $E^i_- := \min_{a\in A} E^i_a$.

We claim that the optimal bid b^i_* for policy i equals $(E^i_+ - E^i_-)/2$, and upon winning the bidding the optimal action is $a_+ = \arg\max_{a\in A} E^i_a$. Notice that no matter whether policy i becomes the winner or the loser, its net reward is at least $(E^i_+ + E^i_-)/2$: if it wins and chooses a_+, after paying the bidding penalty, the net reward is $E^i_+ - (E^i_+ - E^i_-)/2 = (E^i_+ + E^i_-)/2$; if it loses, no matter what action the opponent chooses, its nominal reward is at least E^i_-, and after the bidding reward, the net reward is $E^i_- + (E^i_+ - E^i_-)/2 = (E^i_+ + E^i_-)/2$. If policy i bids $b^i < b^i_*$, then upon losing, its net reward will be $E^i_- + b^i < E^i_- + b^i_* = (E^i_+ + E^i_-)/2$. If it bids $b^i > b^i_*$, then upon winning, its net reward will be $E^i_+ - b^i < E^i_+ - b^i_* = (E^i_+ + E^i_-)/2$. Therefore, the optimal bid is $b^i_* = (E^i_+ - E^i_-)/2$, which is what each selfish policy is expected to select.

Suppose, policy i is the winner. Then, for every $j \neq i$, $b^i_* \geq b^j_*$, i.e., $(E^i_+ - E^i_-)/2 \geq (E^j_+ - E^j_-)/2$. Simplifying, we get $E^i_+ + E^j_- \geq E^i_- + E^j_+$. It follows that $E^i_+ + \sum_{j\neq i} E^j \geq E^i_+ + \sum_{j\neq i} E^j_- \geq E^i_- + E^k_+ + \sum_{j\neq i,k} E^j$ for every for every $k \neq i$. Since the MO-MDP is purely competitive, there will be at least a single k such that a given action is optimal for k, and therefore the claim follows for the single-shot case.

Now, for the general multi-shot case, we inductively apply the above principle in the Bellman equation, which extends the claim to paths of arbitrary length. The convergence of the Bellman iteration is guaranteed because it is a contraction mapping (since $\gamma < 1$). **KM:** I am not sure about this extension. □

4 Multi-Policy Bidding for Multi-Objective RL

We use multi-objective Markov decision processes (MO-MDP) to model uncertain environments and multiple objectives. An MO-MDP with $m \in \mathbb{N}_{>0}$ objectives is specified by a tuple $\mathcal{M} = (S, A, T, R, \gamma, \mu_0)$ where S is the set of states, A is the set of actions, $T : S \times A \times S \to \mathbb{R}$

is the transition function so that $s' \mapsto T(s, a, s')$ forms a probability distribution over S for all $(s, a) \in S \times A$, $\mathbf{R} = \{R_i \colon S \times A \times S \to \mathbb{R}\}_{i \in [1;m]}$ is the set of reward functions corresponding to the m objectives, $\gamma \in (0, 1)$ is the discount factor, and $\mu_0 \in \mathbb{D}(S)$ is the initial state distribution. A *policy* in an MO-MDP $\mathcal{M}$ is a function $\pi : (S \times A)^* \times S \to \mathbb{D}(A)$ that maps a history of state-action pairs and the current state to a distribution over actions. The policy π gives rise to a random sequence of states and actions $s^0, a^0, s^1, a^1, \ldots$, that produces a random sequence of rewards $r_i^0 r_i^1 \ldots$ for each objective $i \in [1; m]$, where $r_i^t = R_i(s^t, a^t, a^{t+1})$ for every $t \in \mathbb{N}$. The *value* of π, denoted $V(\pi)$, is a vector in $\mathbb{R}^m$ such that $V(\pi)_i = \sum_{t=0}^{\infty} r_i^t$. In the multi-objective setting, a commonly studied problem is to find a policy in the pareto-frontier of the value function which balances the tradeoffs between the objectives according to domain requirements (Van Moffaert & Nowé, 2014).

We consider the compositional approach to policy synthesis for MO-MDPs, where we will design a selfish, *local* policy maximizing each individual value component, towards the fulfillment of some required global coordination requirements. The main crux is in the composition process, where each local policy may propose a different action, but the composition must decide one of the actions that will be actually executed. Importantly, the composition must be implementable in a distributed manner, meaning we will *not* use any global policy that would pick an action by analyzing all local policies and their reward functions. **TODO:** running example

We present a novel *auction*-based RL framework for compositional policy synthesis for MO-MDPs. In our framework, not only do the local policies emit actions, but also they *bid* for the privilege of executing their actions for a given number of time steps $\tau \in \mathbb{N}_{>0}$ in future. The bids are all non-negative real numbers, and the highest bidder's actions get executed for the following τ consecutive steps, with bidding ties being resolved uniformly at random. To discourage policies from always bidding high, we penalize the policies proportional to their bids. Thus, each policy can adjust its bid in proportion to the importance for it to execute its action in the current state, and the associated bidding penalty/reward aims to incentivize policies to be truthful. By making the highest bidder active, it is effectively guaranteed that the most important policy is executed. This way, we obtain a purely decentralized scheme to coordinate local policies in a given MO-MDP.

Remark 2 (On the parameter τ). The parameter τ controls how frequently the agent changes its policies. In practice, if τ is too small, the switching could be too frequent for any of the objectives to be fulfilled. For example, **TODO:** running example...

We formalize this approach by viewing the MO-MDP as an m-player general-sum Markov game (**?**) in which m players, one corresponding to each objective, compete with each other to take control of the agent and maximize their objectives. We define the game as follows:

Definition 3 (Markov bidding game)**.** Let $\mathcal{M} = (S, A, T, R, \mu_0)$ be an MO-MDP with m objectives. We define the m-player Markov bidding game as a tuple $\mathcal{B}_{\mathcal{M}} = (\hat{S}, \hat{A}, \hat{T}, P, \hat{R}, \hat{\mu}_0, \tau, \rho)$ where

- $\hat{S} = S \times [1; m] \times [0; \tau]$ is the new state space augmented with the index of a player and the number of remaining timesteps in the range $[0; \tau - 1]$ where $\tau \in \mathbb{N}_{>0}$ is the control horizon. An element of $\hat{S}$ is denoted (s, k, t).
- $\hat{A} = (A \times \mathbb{R}_{>0})^m$ represents the action space of the m players in which each player selects an action from A and a positive bid. We denote a set of actions chosen by the m players with $(\mathbf{a}, \mathbf{b}) = (a_i, b_i)_{i \in [1;m]}$.
- $\hat{T} : \hat{S} \times \hat{A} \times \hat{S} \to \mathbb{R}$ is the new transition function defined as,

$$\hat{T}((s, k, t), (\mathbf{a}, \mathbf{b}), (s', k', t')) \coloneqq \begin{cases} T(s, a_k, s') & k = k' \wedge t' = t - 1 \\ \frac{1}{|\mathbf{b}_{\max}|} T(s, a_{k'}, s') & t = 0 \wedge t' = \tau - 1 \wedge k' \in \mathbf{b}_{\max} \\ 0 & \text{otherwise} \end{cases}$$

where $\mathbf{b}_{\max} \coloneqq \{i \mid b_i = \max\{b_1, \ldots, b_m\}\}$ is the set of agents with maximal bids.

- $\{\hat{R}_i : \hat{S} \times \hat{A} \times \hat{S} \to \mathbb{R}\}_{i \in [1;m]}$ is the set of reward functions for the m agents with

$$\hat{R}_i((s,k,t),(\mathbf{a},\mathbf{b}),(s',k',t')) := \begin{cases} R_i(s,a_k,s') & t > 0 \wedge i = k \\ R_i(s,a_{k'},s') - \rho \cdot b_{k'} & t = 0 \wedge i = k' \end{cases}$$

  where $\rho \in \mathbb{R}_{>0}$ is the bid penalty factor.
- $\hat{\mu}_0 := (\mu_0, 1, 0)$ is the initial state distribution over $\hat{S}$ induced by μ_0 and the second and third components are fixed to be $1, 0$ respectively without loss of generality.

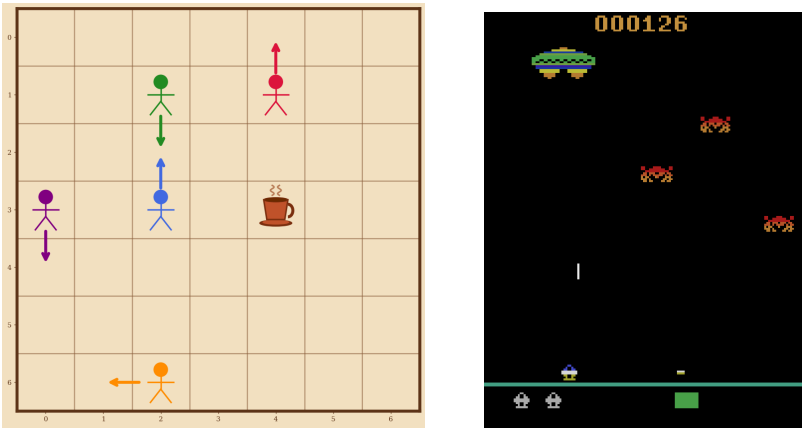
TODO: explain the definition a bit more either before or after

TODO: maybe have a few short comments about literature around multi-agent RL.

5 Environments with Evolving Objectives

We begin by describing the multi-objective environments that we evaluate our framework on. Despite their practical relevance, environments with multiple evolving objectives have received little attention in the multi-objective RL literature.

Coffee Shop. We implement a simple but challenging gridworld environment which consists of an $n \times n$ grid with m targets that are moving at a small rate and stochastically change directions. Each target stays alive for a certain number of timesteps before which the agent has to reach it. Once the target expires or the agent reaches it, another new target immediately respawns at a random position in the grid. The metric the agent has to maximize is the difference between the number of targets reached and the number of targets that expired. This flexible environment also allows us to introduce more targets at deployment time.



(a) Coffee Shop (b) Atari Assault

Figure 1: Preview of the two evaluation environments.

In our experiments, we choose a 30×30 grid with 8 targets. The observations for all the local policies includes information about all the targets (positions and expiry times) and the agent; but they are reordered so that the specific target of a policy is present at the first index. This enables sharing the same actor network across policies. When the agent reaches a target before the expiration, we reward the corresponding policy and penalize it if not. Both the reward and the penalty are equal in magnitude. Furthermore, we include a distance based reward shaping for each policy with respect to its target which is crucial to the learning. **TODO:** Include this as an advantage of our framework in introduction.

Atari Assault. We also test our method on the Atari game Assault in which we control a missile and shoot alien ships that move. Even though this is not a multi-objective environment, we decom-

pose it so that we can treat each alien ship as a different objective. Since these alien ships move around and there are a variable number of ships at any point of time, it is a perfect testbed for our modular approach. We check whether we can synthesize interpretable policies that split control of the missile to target different ships. However, as the standard Atari games from the Arcade Learning Environment (**?**) either use image or raw RAM observations, we cannot decompose the game score and design rewards for the multiple policies. Instead, we turn to the Object-Centric Atari environments (**?**) which exposes object positions and states by decoding RAM states. This permits us to design decomposed observations and rewards for each policy to pursue and shoot the alien ships. However, the metric we optimize is still the game score.

The observation space consists of the positions of the cannon, the alien ships and enemy missiles, current health level, and the number of remaining lives. Similar to Dynamic Targets, we reorder the positions of the alien ships in the observations for each policy. Instead of rewards based on the game score, we design the rewards for each policy. We reward a policy if it successfully destroys an alien ship and penalize it if it loses a life.

6 Implementation

? demonstrate that proximal policy optimization (PPO; **?**) can be effective to train multiple policies concurrently for multi-agent settings. We follow their recipe and train all the local policies with PPO. Similar to **?**, since the objectives in our environments share the same characteristics, we use the same actor and critic network for all the policies. At each iteration of PPO, we collect all the rollouts from each policy's perspective and sample minibatches from this pool to perform the PPO updates.

Additionally, in the Dynamic Targets environments, we employ an attention pooling architecture to encode information about all the targets. This allows each local policy to access information about all targets being pursued by competing policies. It also enables competition with an arbitrary number of policies (i.e., more targets) at deployment time. Each target information vector is passed through an MLP to obtain a target embedding. We then take a weighted sum of these embedding vectors where the weights are obtained by taking the dot product of the embedding vector with a learned query vector. This weighted sum is accessible to all the policies at each step.**TODO:** Talk more about action windows maybe

TODO: explain different bidding mechanisms, only self target visible variants, reward shaping.

6.1 Baselines

We compare our method against two baselines: a single policy PPO with weighted rewards and the deep W-learning (DWN; Rosero et al. (2024)). For the PPO baseline, since all the objectives have the same importance across the two environments, we use the total reward across objectives at each step as the reward function. The observation space remains the same as the multi-policy setting. However, we do not implement any reward shaping for the Dynamic Targets environment due to the difficulty of specifying a meaningful dense reward signal in the presence of multiple targets. We borrow the PPO implementation from CleanRL (**?**). We use the specified PPO hyperparameters from CleanRL for Atari Assault and we tune the hyperparameters for the Dynamic Targets environment with 40 Optuna iterations (**?**).

DWN is another multi-policy method for multi-objective RL that performs DQN for each of the local policies and additionally learns a W-network for each policy that predict scalar weights. These weights indicate which policy's action should be played at each step. The W-network is trained alongside the Q-network by updating it with a gradient step to match the negative TD-error since this indicates how important it is for the policy to play its action. The implementation is borrowed from their repository.

7 Experimental Results

We conduct experiments to show the superior performance of our method against the baselines. We also perform ablations with respect to all the key components of the framework and draw insights on how we can tune them.

7.1 Comparison with Baselines

Dynamic Targets.

Atari Assault.

7.2 Interpretability, Fairness, and Generalizability

Interpretability.

Fairness.

Scaling number of targets.

7.3 Design Choices and Ablations

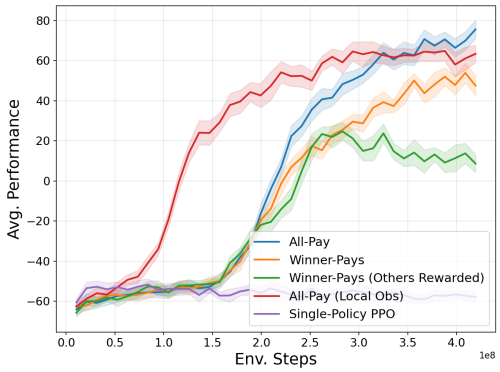
Bid penalty factor.

Bidding range.

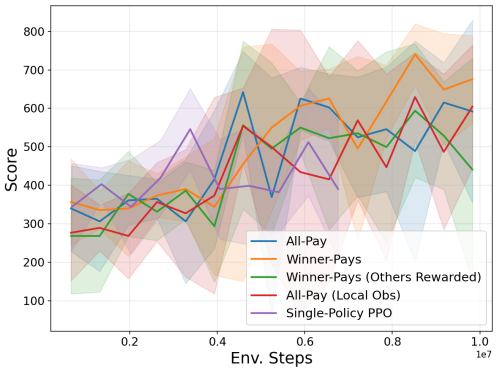
Local vs. global observations. **TODO:** Also talk about attention mechanism here.

Bidding mechanisms.

Action window.



(a) MovingTargetsGridworld. [Placeholder: describe convergence behavior, relative performance of mechanisms, and any notable differences in sample efficiency.]



(b) Atari Assault. [Placeholder: describe convergence behavior, relative performance of mechanisms, and any notable differences in sample efficiency.]

Figure 2: Learning curves for different bidding mechanisms across both environments.

8 Limitations, Conclusion, and Future Directions

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Table 1: Gridworld performance (mean with 95% CI) at the best evaluation iteration. Values are $8 \times$ avg performance.

Algorithm	**Avg. Performance**
All-Pay	75.60 [71.12, 80.08]
Winner-Pays	53.90 [48.84, 58.96]
Winner-Pays (Others Rewarded)	24.75 [20.42, 29.08]
All-Pay (No Reward Shaping)	−40.20 [−43.06, −37.34]
Multi-Agent PPO (Local Obs)	64.85 [59.72, 69.98]
DWN	−41.40 [−45.48, −37.32]
Single-Agent PPO	−51.65 [−54.50, −48.80]
Nearest Target	219.75 [215.81, 223.69]
Least Time Left	89.90 [83.98, 95.82]

Table 2: Assault score (mean with 95% CI) at the best evaluation iteration.

Algorithm	**Score**
All-Pay	642.20 [536.39, 748.01]
Winner-Pays	740.20 [660.53, 819.87]
Winner-Pays (Others Rewarded)	594.00 [420.50, 767.50]
All-Pay (Local Obs)	629.40 [513.95, 744.85]
DWN	294.00 [134.32, 453.68]
Single-Policy PPO	546.00 [440.08, 651.92]

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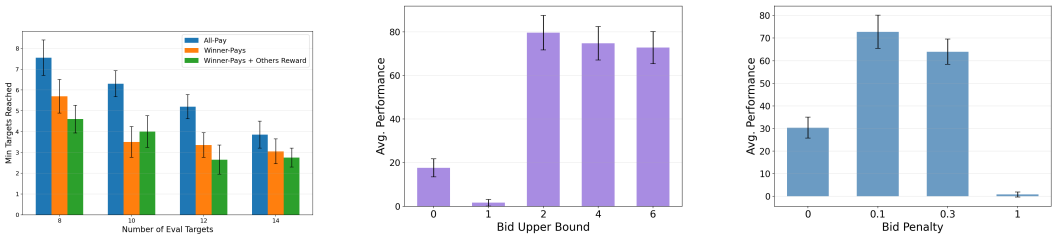
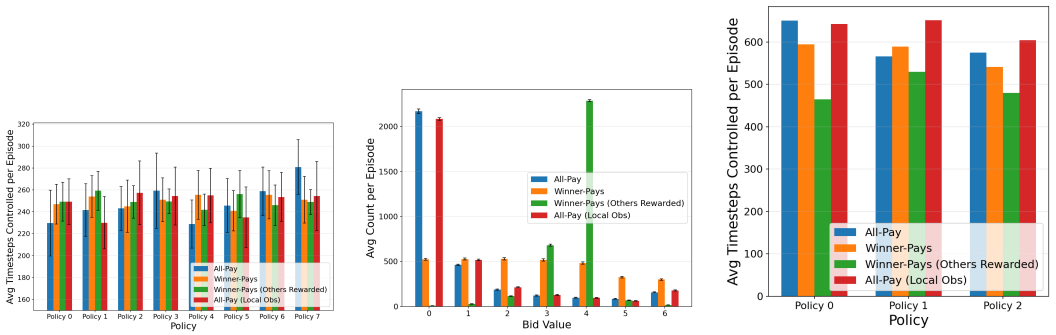


(a) Bidding mechanism scaling. [Placeholder description.]

(b) Bid upper bound. [Placeholder description.]

(c) Bid penalty. [Placeholder description.]

Figure 3: Ablation studies on the MovingTargetsGridworld environment.



(a) MovingTargetsGridworld control timesteps per policy.

(b) Bid distribution per mechanism in MovingTargetsGridworld.

(c) Atari Assault control timesteps per policy.

Figure 4: Comparison of bidding mechanisms: control timesteps per policy in MovingTargetsGridworld (left) and Atari Assault (center), and bid value distributions (right).

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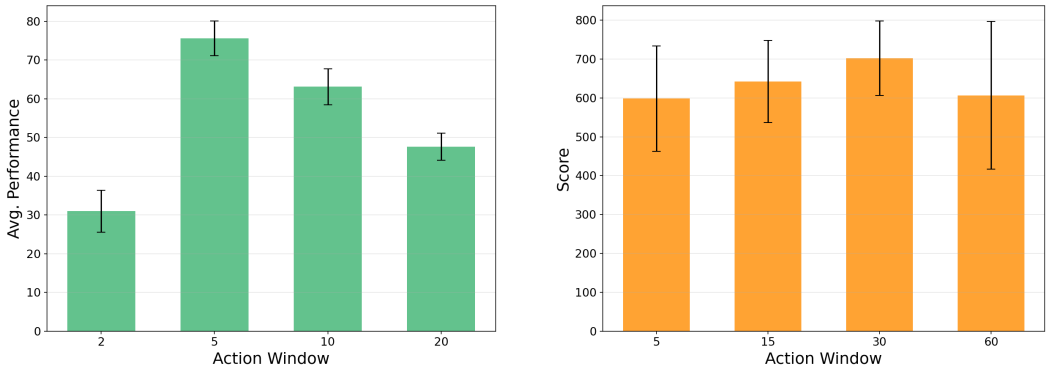

(a) MovingTargetsGridworld.

(b) Atari Assault.

Figure 5: Effect of action window size on performance across both environments.

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